Experiment : 8

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Q | Partition Array for maximum Sum :

Problem definition: Given an integer array arr[]

Partition it into subarrays of length at most K

→ Change every value of subarray to change with their max value

→ Return largest sum of given array after partitioning

Approach :

→ Divide given array into contiguous subarrays whose length is atmost K

→ ~~for~~ for each subarray find maxelement.

→ Replace every element of that subarray with max element

Goal is to maximize final sum.

Solution # code

```
class Solution{
public:
int solve (int i, vector<int>& arr, int k, vector<int>
&dp) {
int n = arr.size()

if (i==n) return 0;

if (dp[i] != -1) return dp[i];
```

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```
int maxi = 0;
int ans = 0;
for (int j = i ; j < min (n, i+k); j++){

      maxi = max(maxi, arr[j]);

   int len = j-i +1;
   int sum  = len* maxi + solve (i+1, arr, k, dp)
      ;

   ans= max (ans, sum);
 }
 return dp[i] = ans;

}

int   maxSumAfterPartitioning (vector<int>& arr , int k){

      int n = arr.size()

   vector<int> dp (n,-1);
   return solve(0, arr, k, dp);

      } };
```